



Does Coding Late at Night Affect Productivity?



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(IT1212)

ABSTRACTION

The research is focused on the impact of coding time morning compared to night on students' productivity. The survey was conducted on the students who code in order to gather information about their study time and self-reported productivity levels. The statistical procedures, including descriptive statistics and hypothesis testing in R, were used to analyze the data. The findings show that there is no variation in productivity among morning and night coders. However, the results shows the significance of personal time management and personal learning style in achieving maximum productivity.

DECLARATION

I hereby declare that this statistical report, "Does Coding Late at Night Affect Productivity?" has been developed in full by myself (B.M.S.K. Basnayaka, SA23800946) between 2025-03-14 and 2025-04-01, with assistance from Mr. Chamith Jayasinghe, as a partial completion of the Probability and Statistics module for the SLIIT CITY UNI BSc (Hons) Computer Science and Software Engineering degree. I further declare that the data included in this study was collected through surveys and is entirely original. I guarantee that the report has not been submitted to any other institution for the award of any degree because it was originally prepared for the "Probability and Statistics" module required by SLIIT CITY UNI.

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LIST OF ABBREVIATIONS

1. SD – Standard deviation
2. Q1 – First quartile
3. Q3 – Third quartile
4. H_0 – Null hypothesis
5. H_1 – Alternative hypothesis

CHAPTER 1 INTRODUCTION

1.1 Background

Coding is a necessary skill for students studying software engineering and other related fields in today's digital age. Students code at various times of day depending on their work habits, lifestyle, and deadlines. But according to research, coding at night may affect one's ability to think properly, sleep, and be productive in general. Some students report that coding in the morning helps them stay cognitively awake and work more effectively, while others feel more focused at night. The goal of this study is to determine whether students who code in the morning and those who code at night are significantly more productive.

1.2 Problem Identification

There is endless debate about the most suitable time for students to do coding tasks. Late-night coding is more comfortable and concentrated for some students, whereas early-morning work is preferred by others. However, not much research has been done on how much time spent coding affects productivity. In this study, we investigate whether students who code at night are less productive than those who code in the morning. Understanding these differences can help students improve their study habits and achieve better learning results.

1.3 Significance of Research

The main objectives of this study are:

1. To compare the productivity scores of students who code in the morning versus those who code at night.
2. To analyze the relationship between coding time and sleep patterns.
3. To determine whether coding at night has a significant impact on efficiency and cognitive performance.

1.4 Objectives of the Study

This research is significant for students, professors, and researchers in software engineering. Understanding the optimal time to code can help students improve their learning experience, increase their productivity, and strike a healthy balance between study and rest. Findings in this research might be beneficial in providing knowledge for teachers and educational institutions. This will make them plan properly for study routines and guidelines as well as assisting students to achieve proper time utilization.

CHAPTER 2 LITERATURE REVIEW

2.1 Introduction

The literature review summarizes current studies of the impact of study time, productivity, and sleeping habits on each other. In this part, we discuss previous studies concerning mental performance and coding effectiveness during different hours of the day. By studying these previous works, we aim to understand how students' ability to concentrate and perform well is affected by coding schedules.

2.2 Study Habits and Productivity

Smith (2021) discovered that students who establish healthy study habits, for example, good time management and constant revision of their assignments, generally perform well academically. Likewise, a review of a study showed a positive relationship between having an organized study routine and academic performance (Research Review International Journal of Multidisciplinary, 2021).

2.3 Effects of Late-Night Work on the Brain

A meta-analysis conducted by Chellappa et al. (2020) showed that night shift work may weaken cognitive function, such as attention and memory. However, a study suggested that those who perceive themselves as night owls would have better cognitive function than those who are early risers (The Guardian, 2024).

2.4 Time Management for Coding Students

Effective time management is extremely crucial for academic success. Planning and not procrastination were the focus of research to increase productivity (International Journal of Humanities Social Science and Management, 2023). Furthermore, using techniques such as the Pomodoro technique has been found to be effective in helping students increase concentration (ERIC, 2023).

2.5 How Sleep Affects Concentration

Getting enough sleep is very important for good cognitive performance. A review of studies showed that sleep problems, such as those caused by working at night, can lead to a decline in cognitive abilities (MDPI, 2021). On the other hand, taking short naps during long periods of work can help improve alertness and performance (Sleep, 2020).

2.6. Summary

The literature identifies different results. There are results that suggest coding late at night may affect cognitive processes, such as memory and attention, negatively. Other results, however, suggest that there may be benefits for those who are naturally night persons and discover they are more productive when working at night. The following chapters of this report will explore these aspects in more depth, especially in the case of students coding.

CHAPTER 3 THEORY & METHODOLOGY

3.1. Research Design

This research adopts a quantitative design as it seeks to gather numerical data in order to compare the correlation between coding hours and productivity. The research process entailed the conduct of a survey, where data was collected from coding students through the use of a questionnaire prepared through Google Forms. The main purpose of this research is to compare students' productivity when coding at night with those coding during the day. It looks at several factors, including:

- The most common times students prefer to code.
- The relationship between the time of coding and self-reported productivity.
- The effect of coding late at night on sleep, focus, and academic performance.

Using both descriptive and inferential statistical analysis, this study is going to explore if coding late at night has a significant effect on a student's efficiency and mental performance.

3.2 Data Collection Method

Data gathering for this study was conducted via an online survey using Google Forms, which was sent to coding students. The survey was designed to gather categorical and numerical data, thus the capacity to perform statistical analysis on the results.

The questionnaire was divided into four sections:

Section 1: Demographic Information

- General information on participants, such as age, gender, study year, and experience with coding.

Section 2: Coding Habits

- This section asks participants what times of day they like to code, for how many hours they code daily, and their work habits, such as whether they take breaks or code constantly without breaks.

Section 3: Productivity Levels

- This item asks respondents to compare their productivity when they code at different times of the day, such as in the morning, afternoon, or evening.

Section 4: Sleep & Health Impact

- This section talks about how late-night coding affects students' sleep quality, their focus, and their overall academic performance.

All the answers were collected anonymously so as to keep the participants' privacy intact and also to make them feel at ease to answer honestly and accurately.

3.3 Structure of the Questionnaire

The questionnaire was designed with the help of multiple-choice, dropdown, and scale-based questions in order to generate accurate and consistent data collection. The structure of the questionnaire outline is given below:

1. Demographic Information:

- Age group
- Gender
- Year of study
- Coding experience level

2. Coding Habits:

- Preferred coding time (Morning, Afternoon, Night, Late Night)
- Hours spent coding per day
- Work habits (Frequent breaks vs. Continuous coding)

3. Productivity Levels:

- Self-rated productivity (1-5 scale)
- Most productive time of the day
- Does coding late at night improve problem-solving?

4. Sleep & Health Impact:

- Average sleep hours
- Feeling tired after late-night coding
- Sleep issues (restlessness, difficulty sleeping)
- Effect on academic performance

This structured format helps organize the data in a way that allows for systematic analysis and comparison of the results.

3.4 Preliminary Data Analysis

Some initial checks on the data were conducted before conducting detailed statistical analysis so that the quality and accuracy of responses that had been gathered could be verified. The following descriptive statistics were used in summarizing data:

3.4.1 Descriptive Analysis

- Mean, Median, and Mode were used in summarizing coding hours, productivity levels, and sleeping hours.
- Frequency Distribution was used in determining the number of students who like coding at different times of the day.
- Standard Deviation was employed to understand how far individual responses varied from the mean.

3.1.1 Analyzing Demographic Data

The dataset was categorized into different groups based on factors such as age, gender, and coding experience levels. To visualize the trends in demographic data, bar charts and pie charts were used to make the information clearer and easier to understand.

CHAPTER 4 RESULTS

This chapter presents the findings of the survey of coding students in order to compare their productivity levels based on their coding habits, preferred time of coding, and sleeping habits. The results include demographic distribution, coding habits, productivity levels, and the influence of sleep and health on coding productivity. (The survey gathered responses from 61 participants.)

4.1 Descriptive Data Analysis

4.1.1 Demographic Information

Age Distribution

- Pie chart – Age

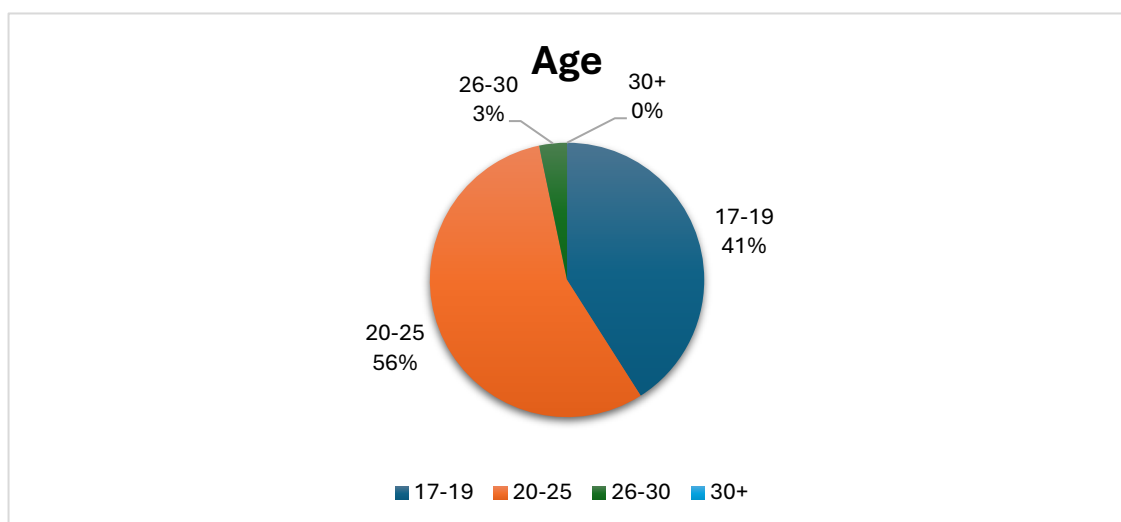


Figure 1: Age

- Table – Age

Age			
17-19	20-25	26-30	30+
25	34	2	0

Table 1: Age

The majority of respondents (56%) are between the ages of 20-25, followed by 41% in the age bracket 17-19. A minority (3%) belong to the 26-30 age group, and there were no respondents above age 30.

- Code – Age distribution

```
R ▾ R 4.4.3 · C:/Users/sandu/OneDrive/Desktop/PS/ ↗  
> data$age <- as.numeric(data$age)  
> summary(data$age)  
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.   
  1.000  1.000   2.000   1.623  2.000   3.000   
> sd(data$age) # Standard deviation  
[1] 0.5526884  
> |
```

Figure 2: Age Distribution

This code analyzes the age column in the data dataset. The `summary(data$age)` command shows key statistics: the smallest age (Min) is 1, the largest age (Max) is 3, and the average (Mean) is about 1.62. The median (middle value) is 2, and the 1st and 3rd quartiles (which divide the data into four parts) are 1 and 2. The `sd(data$age)` command calculates the standard deviation, which is about 0.55. This tells us how much the ages vary from the average.

Gender Distribution

- Pie chart – Gender

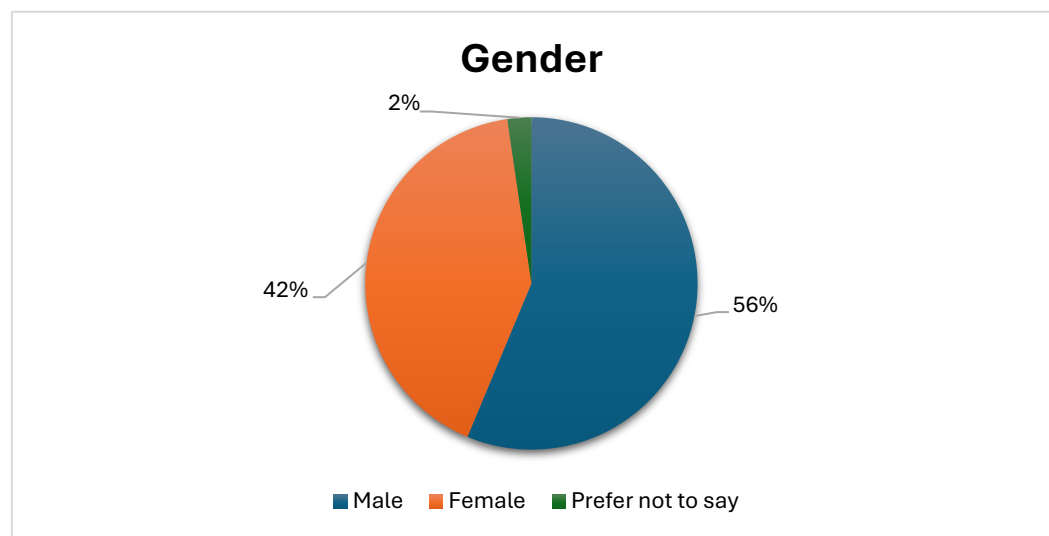


Figure 3: Gender

- Table – Gender

Gender		
Male	Female	Prefer not to say
34	25	2

Table 2: Gender

Among the participants, most were female (56%), while 41% were male, and 3% chose not to disclose their gender.

Year of Study

- Pie chart – Year of Study

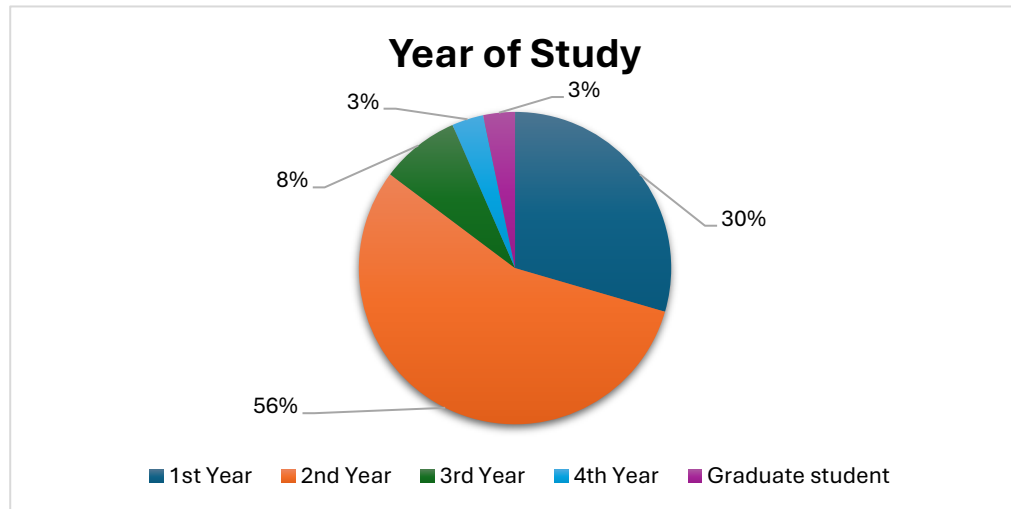


Figure 4: Year of Study

- Table – Year of Study

1 st Year	2 nd Year	3 rd Year	4 th Year	Graduate student
18	34	5	2	2

Table 3: Year of Study

Most of the students who responded were in their 2nd year (56%), while 30% were in their 1st year. A smaller number were in their 3rd year (8%), 4th year (3%), or at the graduate level (3%).

Coding Experience Level

- Pie chart – Coding Experience Level

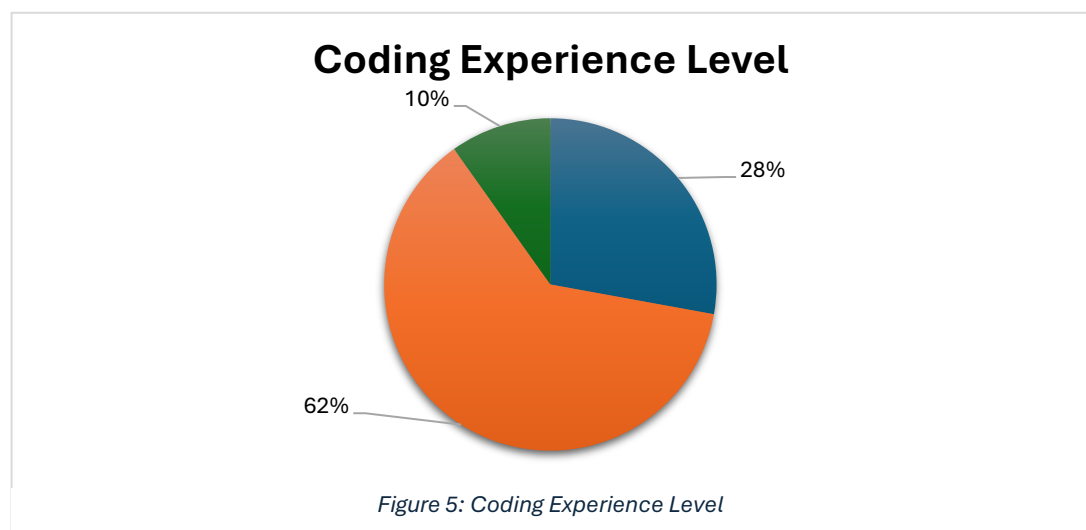


Figure 5: Coding Experience Level

- Table – Coding Experience Level

Beginner	Intermediate	Advanced
17	38	6

Table 4: Coding Experience Level

Most of the students who responded were in their 2nd year (56%), while 30% were in their 1st year. A smaller number were in their 3rd year (8%), 4th year (3%), or at the graduate level (3%).

4.1.2 Coding Habits

Preferred Coding Time

- Pie chart – Coding Hour

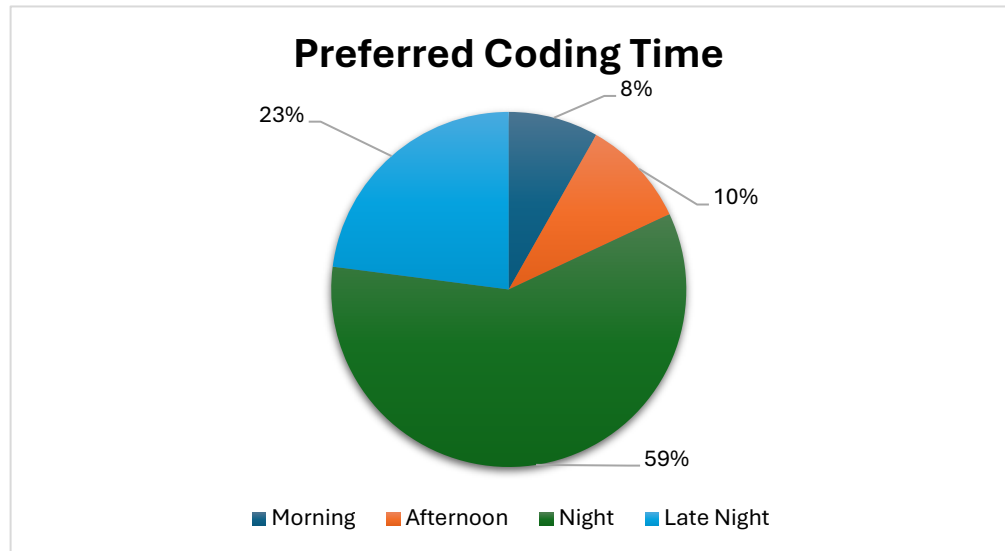


Figure 6: Coding Hour

- Table – Coding Time

Preferred Coding Time			
Morning	Afternoon	Night	Late Night
5	6	36	14

Table 5: Coding Hour

Most students like to code at night, with 59% choosing to do so between 6 PM and 12 AM. About 23% prefer coding even later, from 12 AM to 6 AM. Only a small number of students code in the morning (8.2%) or in the afternoon (9.8%).

Coding Hours per Day

- Pie chart – Coding Hours Per Day

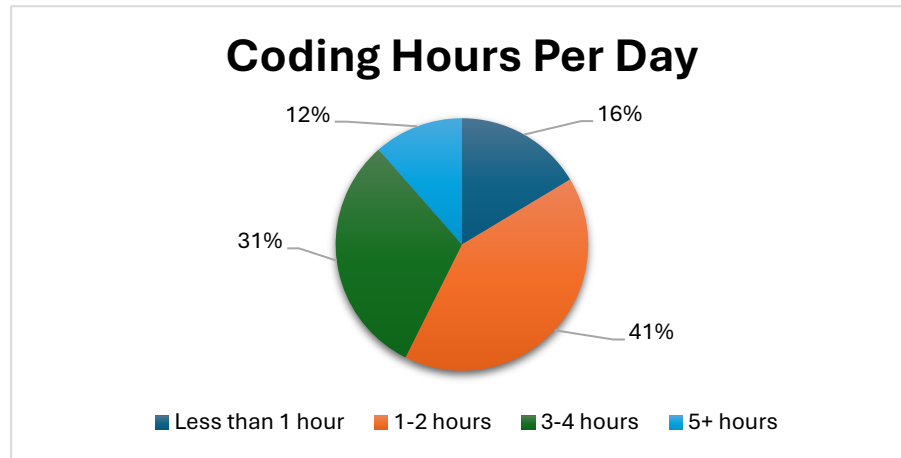


Figure 7: Coding Hours per Day

- Table – Coding Hours

Coding Hours			
Less than 1 hour	1-2 hours	3-4 hours	5+ hours
10	25	19	7

Table 6: Coding hours

Most students spend 1-2 hours coding each day (41%). Around 31% code for 3-4 hours, while 16% code for less than an hour. Only 11% spend more than 5 hours coding daily.

- Code – Descriptive Statistics for Coding Hours

```

Console Terminal x Background Jobs x
R 4.4.3 · C:/Users/sandu/OneDrive/Desktop/PS/
> summary(data$coding_hours)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000  2.000  4.000  3.164  4.000  4.000
> mean(data$coding_hours)
[1] 3.163934
> median(data$coding_hours)
[1] 4
> sd(data$coding_hours)
[1] 1.098185
>

```

Figure 8: Descriptive Statistics for Coding Hours

This code analyzes the “coding_hours” column in the “data” dataset. The “summary(data\$coding_hours)” command gives key statistics: the minimum coding hours are 1, the maximum is 4, and the average (Mean) is about 3.16. The median (middle value) is 4, meaning half the values are below 4 and half are above. The 1st quartile is 2, and the 3rd quartile is 4, showing the data spread. The “sd(data\$coding_hours)” command calculates the standard deviation, which is about 1.10, indicating how much coding hours vary from the average.

Coding Environment and Breaks

- Pie chart – Coding Environment and Breaks

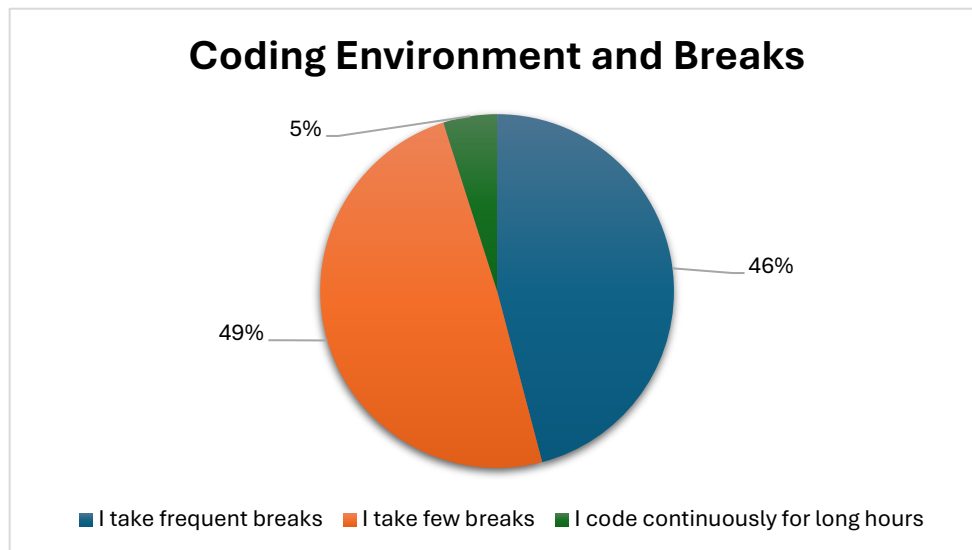


Figure 9: Coding Environment and Breaks

- Table – Coding Environment and Breaks

I take frequent breaks	I take few breaks	I code continuously for long hours
28	30	3

Table 7: Coding Environment and Breaks

When asked about their coding habits, 49% of students said they took a few breaks, while 46% took frequent breaks. Only 5% mentioned coding for long hours without taking breaks.

4.1.3 Productivity Levels

Self-Reported Productivity Scores

- Bar chart – Productivity Scores

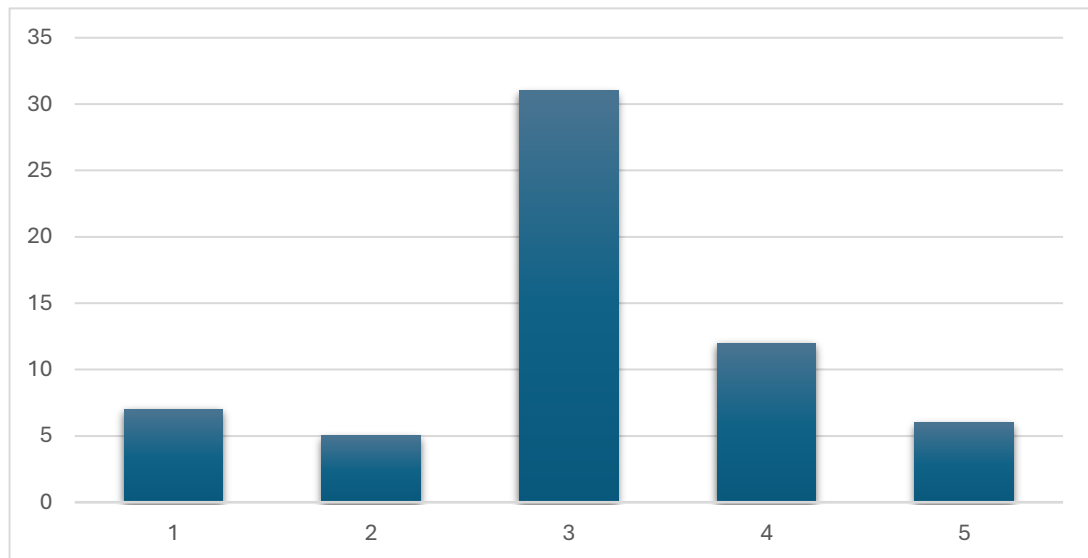


Figure 10: Productivity Scores

- Table – Productivity Scores

1	2	3	4	5
7	5	31	12	6

Table 8: Productivity Scores:

- Code – Productivity distribution

```

Console Terminal x Background Jobs x
R 4.4.3 · C:/Users/sandu/OneDrive/Desktop/PS/
> summary(data$productivity_score)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
 1.000  3.000   3.000   3.082  4.000   5.000
> mean(data$productivity_score)
[1] 3.081967
> median(data$productivity_score)
[1] 3
> sd(data$productivity_score)
[1] 1.069191
> |

```

Figure 11: Productivity Scores Code

This code analyzes the `productivity_score` column in the data dataset. The “`summary(data$productivity_score)`” command shows key statistics: the minimum score is 1, the maximum is 5, and the average (Mean) is about 3.08. The median (middle value) is 3, meaning half the values are below 3 and half are above. The 1st quartile is 3, and the 3rd quartile is 4, showing the data spread. The “`sd(data$productivity_score)`” command calculates the standard deviation, which is about 1.07, indicating how much productivity scores vary from the average.

Most Productive Time

- Pie chart – Most Productive Time

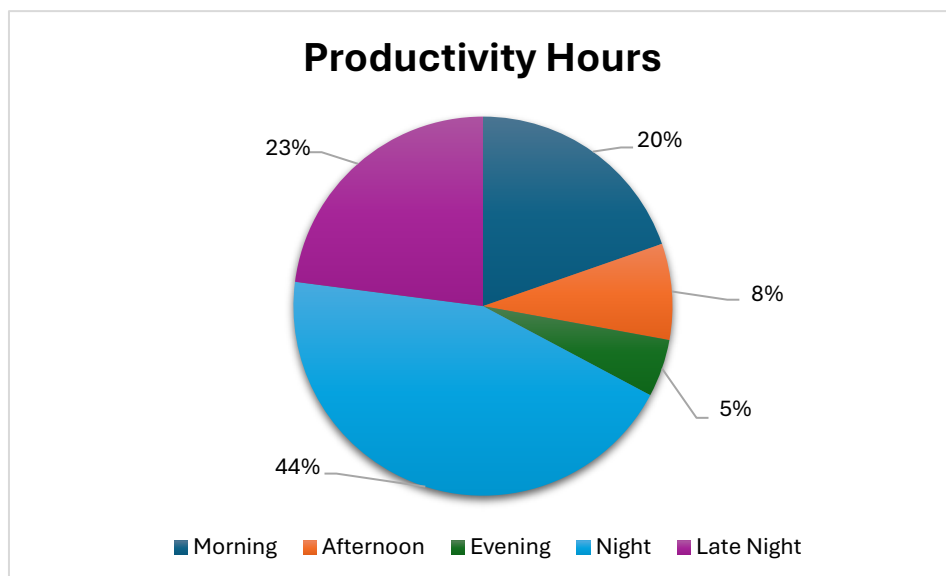


Figure 12: Most Productive Time

- Table – Productivity Hours

Morning	Afternoon	Evening	Night	Late Night
12	5	3	27	14

Table 9: Productive Time

Most students are most productive at night (44%), followed by late-night (23%). Fewer students find the morning (20%), afternoon (8%), or evening (5%) to be their most productive time.

Concentration Levels

- Pie chart – Concentration Levels

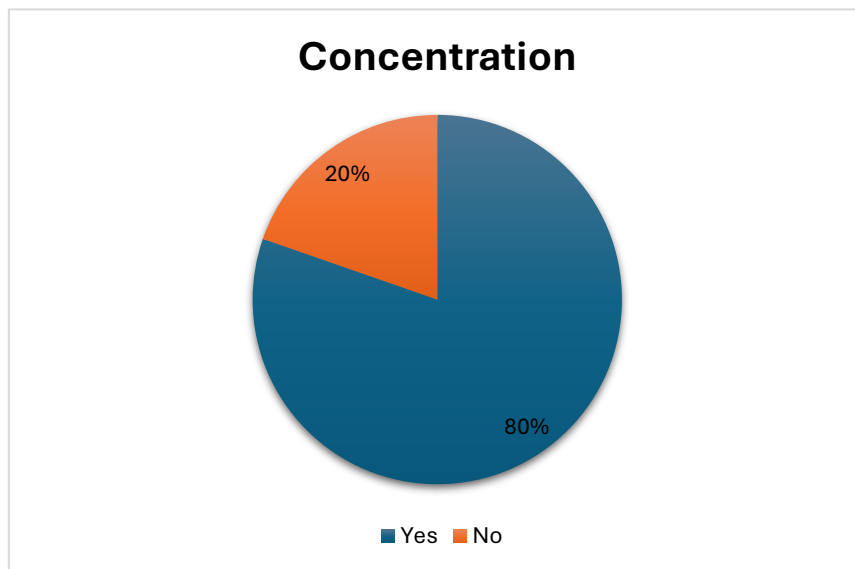


Figure 13: Concentration Levels

- Table – Concentration Levels

Yes	No
49	12

Table 10: Concentration Levels

Most students (80%) said they can focus while coding. However, 20% struggle to stay concentrated. This shows that while most find it easy, some still face challenges.

4.1.4 Sleep & Health Impact

Sleep Duration

- Pie chart – Sleep Duration

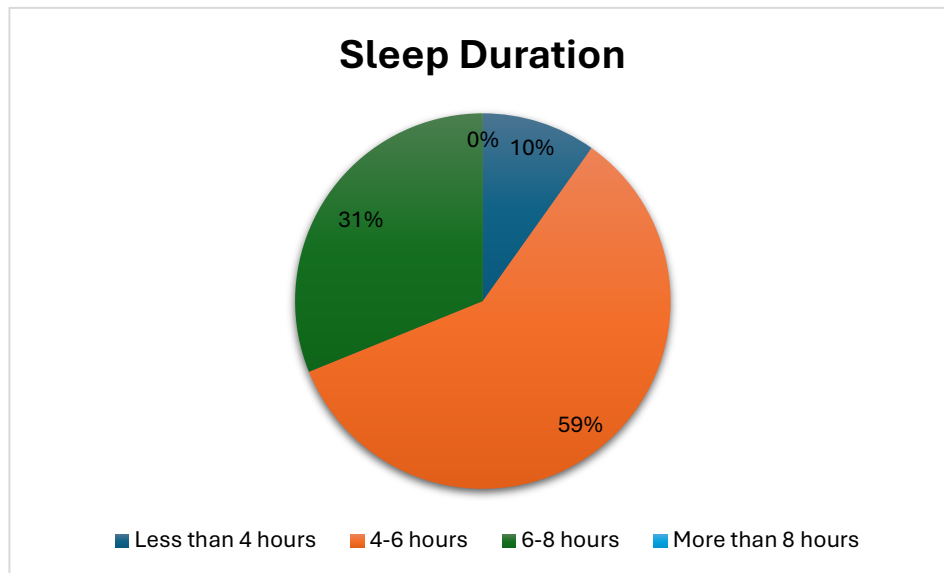


Figure 14: Sleep Duration

- Table – Sleep Duration

Less than 4 hours	4-6 hours	6-8 hours	More than 8 hours
6	36	19	0

Table 11: Sleep Duration

- Code – Descriptive Statistics for Sleep Duration

```

Console Terminal x Background Jobs x
R 4.4.3 · C:/Users/sandu/OneDrive/Desktop/PS/
> summary(data$sleep_hours)
  Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  1.000  1.000   1.000   1.508  2.000   3.000
> mean(data$sleep_hours)
[1] 1.508197
> median(data$sleep_hours)
[1] 1
> sd(data$sleep_hours)
[1] 0.6738682
>

```

Figure 15: Descriptive Statistics for Sleep Duration

This code analyzes the “sleep_hours” column in the “data” dataset. The “summary(data\$sleep_hours)” command shows that the minimum sleep hours recorded is 1, the maximum is 3, and the average (Mean) is about 1.51. The median (middle value) is 1, meaning most students reported around 1 hour of sleep. The 1st quartile is 1, and the 3rd quartile is 2, showing that most values fall within this range. The “sd(data\$sleep_hours)” command calculates the standard deviation, which is about 0.67, indicating how much sleep hours vary from the average. Additionally, the survey data shows that 59% of students reported sleeping 4-6 hours, 31% slept 6-8 hours, and 10% had less than 4 hours of sleep. No students reported sleeping more than 8 hours.

Tiredness and Focus

- Pie chart – Tiredness and Focus



Figure 16: Tiredness and Focus

- Table – Tiredness and Focus

Yes	No
26	35

Table 12: Tiredness and Focus

After coding late at night, some students felt tired or had trouble focusing the next day. About 43% of them experienced these problems, while 57% did not. This shows that late-night coding affects some students more than others.

Sleep Issues

- Pie chart – Sleep Issues

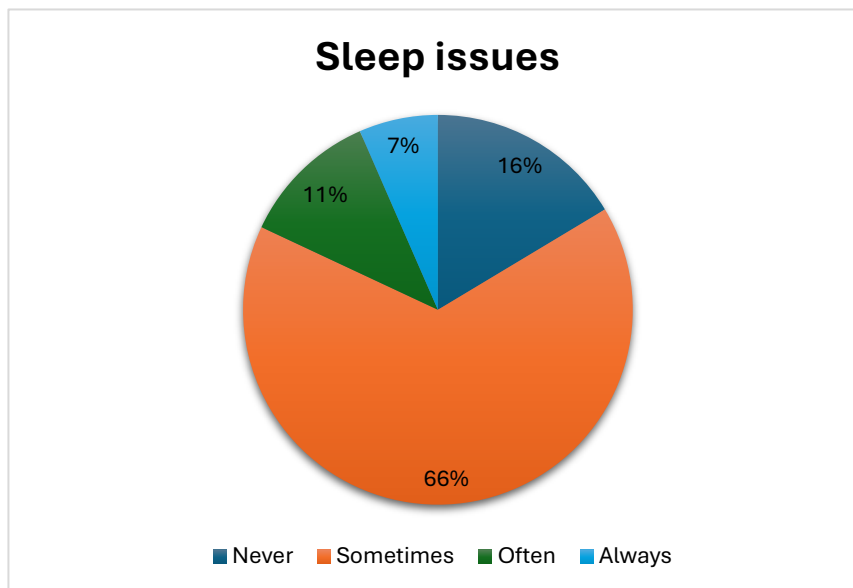


Figure 17: Sleep Issues

- Table – Sleep Issues

Never	Sometimes	Often	Always
10	40	7	4

Table 13: Sleep Issues

Many students sometimes have trouble sleeping (66%), while some never have sleep problems (16%). Only a few students often struggle with sleep (11%), and even fewer face constant sleep issues (6%). This shows that sleep problems are common but vary among students.

Impact on Academic Performance

- Pie chart – Impact on Academic Performance

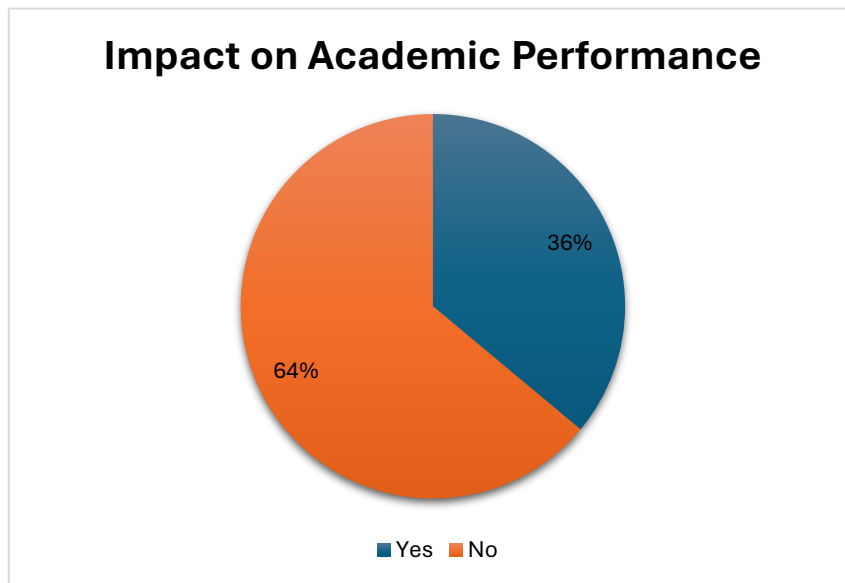


Figure 18: Impact on Academic Performance

- Table – Impact on Academic Performance

Yes	No
22	39

Table 14: Impact on Academic Performance

Most students (64%) think that staying up late to code does not hurt their grades. However, 36% believe it does affect their academic performance. This shows that opinions on late-night coding are divided among students.

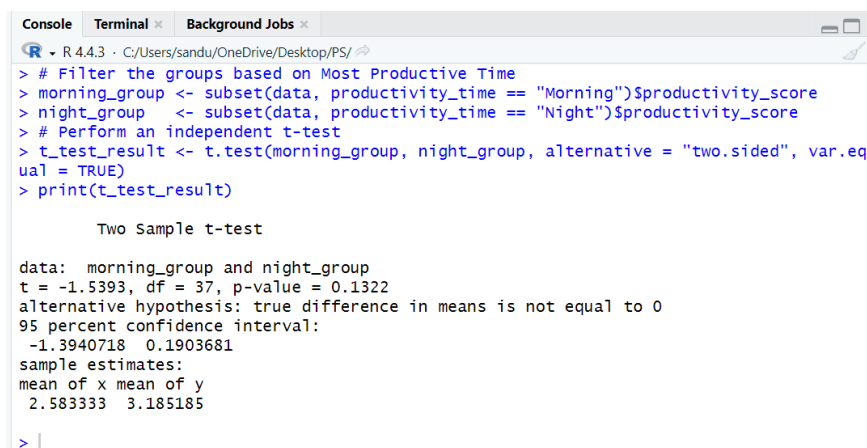
4.2 Statistical Comparisons

4.2.1 t-Test Hypothesis (Productivity Comparison)

We use a t-test to compare morning coders vs. night coders based on their self-reported productivity scores.

- Null Hypothesis (H_0):
There is no significant difference in productivity scores between students who code in the morning and those who code at night.
- Alternative Hypothesis (H_1):
There is a significant difference in productivity scores between students who code in the morning and those who code at night.

Code – t-Test



```
R 4.4.3 · C:/Users/sandu/OneDrive/Desktop/PS/
> # Filter the groups based on Most Productive Time
> morning_group <- subset(data, productivity_time == "Morning")$productivity_score
> night_group <- subset(data, productivity_time == "Night")$productivity_score
> # Perform an independent t-test
> t_test_result <- t.test(morning_group, night_group, alternative = "two.sided", var.equal = TRUE)
> print(t_test_result)

Two Sample t-test

data: morning_group and night_group
t = -1.5393, df = 37, p-value = 0.1322
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -1.3940718  0.1903681
sample estimates:
mean of x mean of y
 2.583333  3.185185
> |
```

Figure 19: t-Test

This code checks if there is a significant difference between the productivity scores of students who are most productive in the morning and those who are most productive at night.

- The “morning_group” contains the “productivity_score” of students who are most productive in the morning.
- The “night_group” contains the “productivity_score” of students who are most productive at night.

t = -1.5393

This is the t-score, which helps determine the difference between the two groups.

df = 37

Degrees of freedom, related to sample size.

p-value = 0.1322

Since this is greater than 0.05, we do not have enough evidence to say that the two groups are significantly different.

Confidence Interval (-1.39 to 0.19)

The true difference in means is likely between these values. Since this range includes 0, it suggests there is no strong difference.

Conclusion

The t-test shows no significant difference in productivity scores between morning and night people. This means that, based on the data, being a morning or night person does not strongly impact productivity.

4.2.2 Chi-Square Test

4.2.2.1 Coding Time vs. Productivity Level

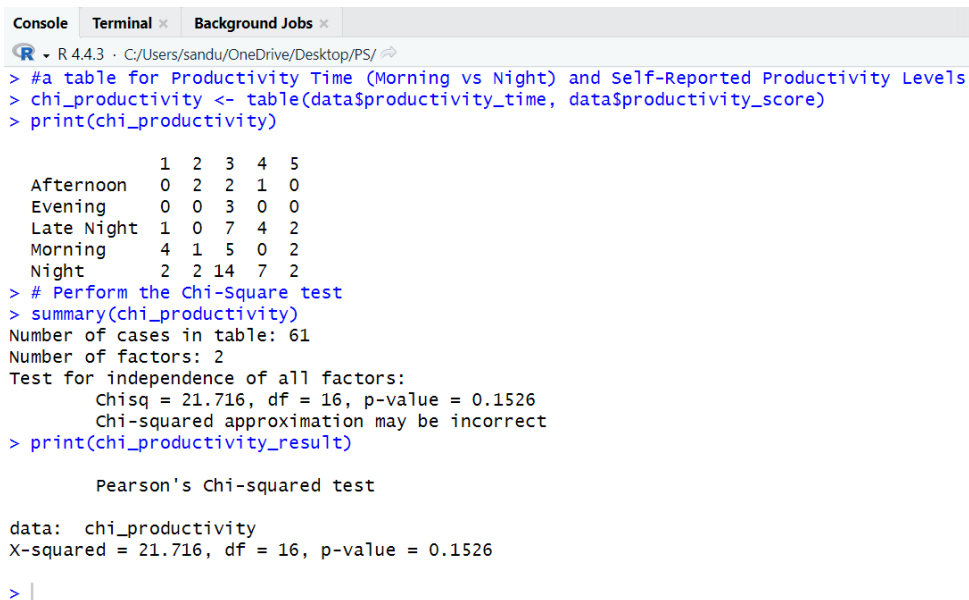
- Null Hypothesis (H_0):

There is no association between Preferred Coding Time (Morning vs. Night) and Productivity Levels. (They are independent.)

- Alternative Hypothesis (H_1):

There is an association between Preferred Coding Time and Productivity Levels. (They are not independent.)

Code – Chi-Square Test 1



```
> #a table for Productivity Time (Morning vs Night) and Self-Reported Productivity Levels
> chi_productivity <- table(data$productivity_time, data$productivity_score)
> print(chi_productivity)

      1  2  3  4  5
Afternoon 0  2  2  1  0
Evening   0  0  3  0  0
Late Night 1  0  7  4  2
Morning   4  1  5  0  2
Night     2  2 14  7  2
> # Perform the Chi-Square test
> summary(chi_productivity)
Number of cases in table: 61
Number of factors: 2
Test for independence of all factors:
  Chisq = 21.716, df = 16, p-value = 0.1526
  Chi-squared approximation may be incorrect
> print(chi_productivity_result)

Pearson's Chi-squared test

data:  chi_productivity
X-squared = 21.716, df = 16, p-value = 0.1526
> |
```

Figure 20: Code – Chi-Square Test 1

The Chi-Square test (`chisq.test(chi_productivity)`) checks if there is a significant relationship between productivity time and productivity scores.

X-squared value = 21.716

This measures the difference between expected and actual values in the table. A higher value usually means a stronger relationship.

Degrees of Freedom (df) = 16

This is based on the number of rows and columns in the table.

p-value = 0.1526

The p-value tells us whether the relationship is statistically significant.

- If **p-value < 0.05**, there is a strong relationship.
- Here, **p-value = 0.1526**, which is greater than 0.05. This means there is no strong evidence that productivity time and productivity score are related.

Conclusion

Based on the p-value, we cannot say that the time students feel most productive affects their productivity score.

4.2.2.2. Coding Time vs. Sleep Issues

- Null Hypothesis (H_0):

There is no association between Preferred Coding Time (Morning vs. Night) and Sleep Issues. (They are independent.)

- Alternative Hypothesis (H_1):

There is an association between Preferred Coding Time and Sleep Issues. (They are not independent.)

Code – Chi-Square Test 2

```
Console Terminal x Background Jobs x
R 4.4.3 · C:/Users/sandu/OneDrive/Desktop/PS/
> print(chi_sleep_issues)

      Always Never Often Sometimes
Afternoon     0     1     0         4
Evening       0     0     0         3
Late Night    1     2     2         9
Morning       2     4     1         5
Night         1     3     4        19
> summary(chi_sleep_issues)
Number of cases in table: 61
Number of factors: 2
Test for independence of all factors:
  Chisq = 9.092, df = 12, p-value = 0.695
  Chi-squared approximation may be incorrect
> print(chi_sleep_issues_result)

      Pearson's Chi-squared test

data:  chi_sleep_issues
X-squared = 9.0921, df = 12, p-value = 0.695
> |
```

Figure 21: Code – Chi-Square Test 2

The table represents the relationship between time of day (Morning, Night, etc.) and self-reported productivity levels (Always, Never, Often, Sometimes). Each number shows how many people fall into each category.

Chi-square value (X^2) = 21.716

Measures how different the observed data is from expected data

Degrees of freedom (df) = 16

Represents how many categories can vary

p-value = 0.1526

The relationship is significant or not

Conclusion

The p-value (0.1526) is greater than 0.05, which means we fail to reject the null hypothesis. In simple terms, the time of day does not significantly affect self-reported productivity levels based on this data.

CHAPTER 5 – DISCUSSION AND CONCLUSIONS

4.1 Discussion

This chapter describes the key survey findings on the productivity of coding students, their coding habits, and how sleep and health influence them. Analysis provides insight into how these factors decide students' coding efficiency.

5.1.1 Demographic Influence

The majority of participants were between 17-25 years of age, and most were in their second year of study. This suggests that the survey encompasses students who are still acquiring the ability to code. Gender distribution was quite level, with slightly more male participants.

5.1.2 Coding Habits and Productivity

The results indicate that most students like to code late at night (59%) or during nighttime (23%). This is also reflected in the productivity scores, where 44% of the students reported most productive during nighttime, followed by 23% during late night. This may mean that the students may perceive the quiet nighttime environment as more favorable for focused work. However, a few said they coded in the morning only (8.2%) or afternoon only (9.8%), perhaps meaning that day-time distractions or school schedules influence their coding schedules.

5.1.3 Time Spent Coding

Most students code between 1-2 hours a day (41%), while 31% code between 3-4 hours. Only 11% code for over 5 hours, which implies long coding periods are not common. This shows that while students do practice coding, they don't continue working on it for extended periods of time due to likely study workload and other commitments.

5.1.4 Breaks and Concentration

Nearly half (49%) of the students have some breaks while coding, and 46% of them have breaks regularly. Only 5% code for hours without taking any break. This justifies that students have a balanced approach to coding by having breaks which might keep them fresh and not sleepy. Moreover, 80% of the students reported good concentrations while coding, which also indicates that the majority of the students control their concentration very well.

5.1.5 Sleep and Its Impact on Productivity

Most students (59%) sleep between 4-6 hours each night, and just 31% enjoy 6-8 hours of sleep. Nobody reported sleeping more than 8 hours. This low level of sleep could affect students' energy and productivity. In addition, 43% of students said that they experienced tiredness or lack of concentration after late-night coding. This finding shows that while night coding is common, it damages alertness the next day.

5.1.6 Academic Performance

The survey also examined the impact of late-night coding on academic grades. While 64% of the students pointed out that it had no impact on their grades, 36% thought that it had a negative impact. The split opinion shows that while some students can manage time effectively, others may not be in a position to multitask between coding and academic work.

5.1.7 Statistical Analysis Findings

A t-test was applied to determine whether the productivity score was different for morning coders compared to night coders. The result supported the null hypothesis and indicated that there was no notable difference between the two groups. This suggests that productivity does not necessarily depend on coding time alone but may be determined by preference and habit.

5.2 Conclusions

The findings show that students coding late at night, having effective breaks, and effective concentration are the patterns of coding students. Sleep deprivation is a cause for worry as most students sleep fewer hours than they should. Though some students feel tired and perform poorly academically due to late-night coding, others manage it. The findings of the t-test show that productivity does not so much depend on the day but work habits.

5.2.1 Recommendations

- **Time Management Techniques:** Students should mix coding with sufficient sleep to prevent fatigue and maintain productivity.
- **Health Considerations:** Encouraging students to have healthier sleeping habits can enhance focus and productivity in academics.
- **Further Research:** The long-term effects of night coding on academic performance and overall well-being may be examined in subsequent research.

In conclusion, this research provides an understanding of students' coding culture and their effects on productivity. While night coding is the trend, students should consider the impact their lifestyle has on their health and academic performance.

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